ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE F · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice f.

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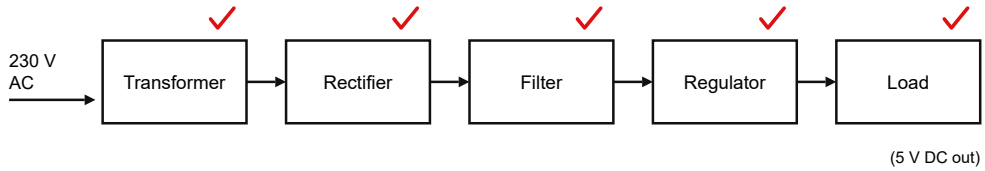
How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers and wording. Circuits may be drawn in different but correct ways: mark each required element and correct connection.

QUESTION 1

1.1 Any THREE sensors ✓ with what they detect ✓, e.g.: inductive proximity sensor – metal objects nearby; capacitive proximity sensor – almost any material, including liquids; photoelectric sensor – objects breaking or reflecting a light beam; limit switch – the end position of a moving part; thermocouple – temperature; pressure switch / transducer – hydraulic or pneumatic pressure; encoder – shaft position or speed. (6)

1.2 A small current through the relay coil ✓ makes an electromagnet ✓ that pulls the armature and closes (or opens) the contacts ✓; the contacts are in a separate circuit, so they can switch a much larger current / voltage while the two circuits stay electrically isolated ✓. (4)

1.3



(Blocks in the correct order with arrows: AC supply → transformer 1 → rectifier 1 → filter 1 → regulator 1 → load / DC output 1) (5)

1.4 It charges at the peaks of the rectified voltage and discharges between them ✓, smoothing the pulsating DC into a steadier DC voltage (reducing ripple) ✓. (2)

1.5 $f = 1 \div T$ ✓ $= 1 \div 0{,}02$ ✓ $= 50$ Hz ✓ (3)

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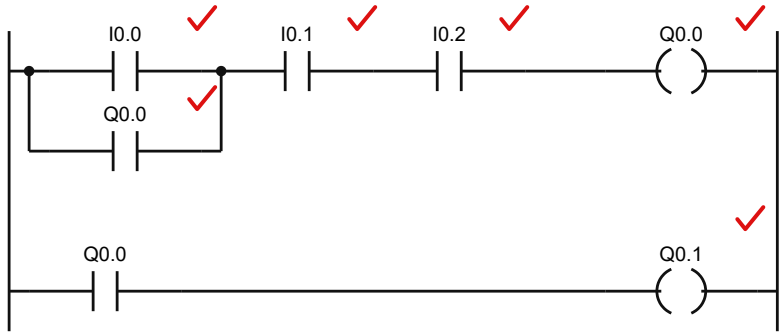
QUESTION 2

2.1 A device that changes an energy signal (electrical, hydraulic or pneumatic) into movement or force ✓✓. Any TWO ✓✓: electric motor; solenoid; hydraulic or pneumatic cylinder; servo motor; stepper motor; relay. (4)

2.2 Swap (interchange) any TWO of the three supply lines to the motor ✓, which reverses the direction of the rotating magnetic field ✓. (2)

2.3 Any FOUR ✓✓✓✓: power supply; CPU (processor); memory; input module; output module; programming device / communication port. (4)

2.4



I0.1 and I0.2 are wired NC, so they are programmed as NO contacts that are closed in normal operation

(I0.0 NO 1 with a Q0.0 holding contact in parallel 1; I0.1 and I0.2 in series, programmed so they are closed when the NC buttons are not pressed 2; output coil Q0.0 1; second rung: Q0.0 contact driving Q0.1 1) (6)

2.5 The PLC repeats a cycle continuously ✓: it reads the state of all inputs ✓, executes the program from top to bottom ✓, and then updates all the outputs ✓ (and does internal checks / communication). (4)

[20]

QUESTION 3

3.1.1 The resistance of a fluid to flow (how thick or thin it is) ✓ (1)

3.1.2 The force acting on a unit area (P = F ÷ A, in pascal) ✓ (1)

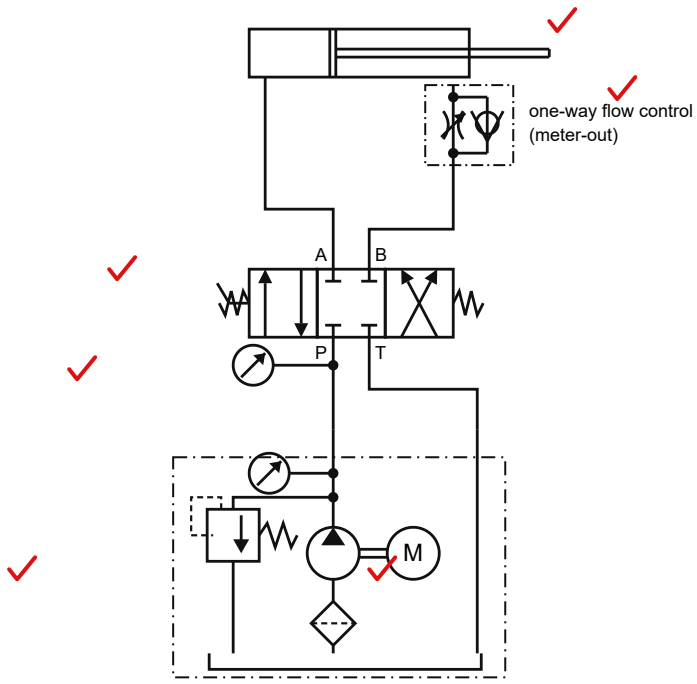
3.1.3 The volume of fluid passing a point per unit of time (e.g. litres per minute) ✓ (1)

3.2 Any FOUR parts ✓ with function ✓: reservoir (tank) – stores and cools the oil and lets dirt and air settle; pump – creates the flow of oil; electric motor – drives the pump; pressure relief valve – limits the maximum pressure and protects the system; filter – removes dirt from the oil; pressure gauge – shows the system pressure. (8)

3.3 It has a check valve in parallel with an adjustable throttle ✓; in one direction the oil flows freely through the check valve ✓, and in the other it must pass through the throttle, which limits the flow rate and therefore the speed of the cylinder ✓. (3)

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3.4



(Power pack (tank, pump, motor, filter) 1; relief valve to tank 1 and pressure gauge on the pressure line 1; 4/3 lever valve with a closed centre 1; double-acting cylinder correctly connected 1; one-way flow control in the rod-end (outlet) line with the check valve allowing free flow into the rod end, so the oil leaving during extension is throttled (meter-out) 1)

(6)

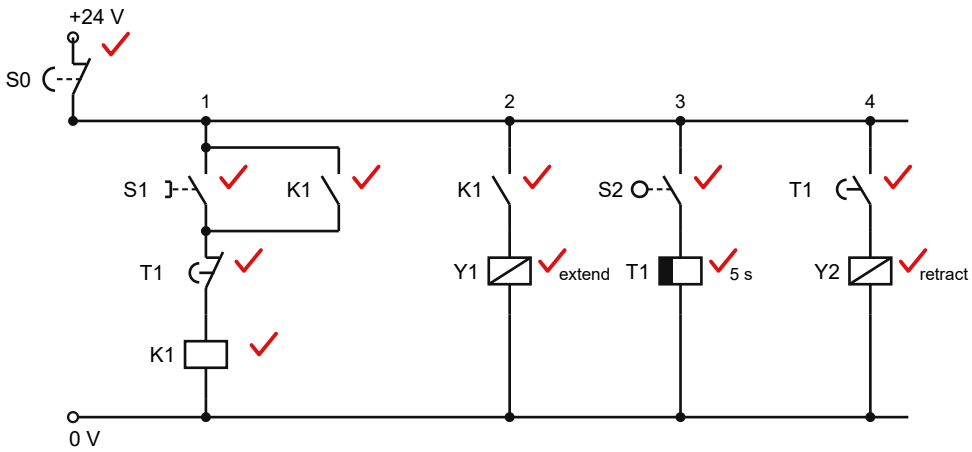
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QUESTION 4

4.1 It detects that a moving part (e.g. the cylinder piston rod) has reached a set position ✓ and sends a signal to the control circuit to start the next step or stop the movement ✓. (2)

4.2 When its coil is energised, it waits for the set time ✓ before its contacts change over ✓; when the coil is de-energised the contacts return immediately ✓. (3)

4.3



(Supply lines 1; E-stop (NC) in the main supply 1; S1 (NO) 1 in series with an NC T1 contact 1 driving relay K1 1 with a K1 holding contact 1; K1 contact switching Y1 1; S2 (NO, actuated at full extension) driving timer T1 set to 5 s 1; T1 (NO) contact switching Y2 1; Y1 and Y2 never energised together 1; correct symbols and neat 1)

(11)

4.4 Any FOUR ✓✓✓✓: the cylinder can be controlled from a distance; easy automatic sequencing with timers, counters and sensors; can be connected to PLCs and computers; safety interlocks and emergency stops are easy to add; faster, repeatable operation; less effort for the operator; fault indication is possible. (4)

[20]

QUESTION 5

5.1 Preventive maintenance is planned and done at regular intervals ✓ to prevent breakdowns (e.g. changing filters and oil, checking hoses) ✓. Breakdown maintenance is done only after a machine has failed ✓, which leads to unplanned downtime and often higher costs ✓. (4)

5.2 Any SEVEN steps in a logical order ✓✓✓✓✓✓✓: switch off the pump and lock out / tag the isolator; lower or support any loads; release all pressure from the system; place a drip tray and cloths under the hose; loosen and remove the hose fittings and plug the open ports to keep dirt out; fit a new hose of the correct size, type and pressure rating, without twisting it; tighten the fittings to the correct torque; top up the oil, bleed air, switch on and check for leaks at low pressure; clean up spilled oil and remove the lock-out. (7)

5.3 Any FOUR ✓✓✓✓: no control voltage (blown fuse, tripped breaker, E-stop pressed); faulty START button or relay contact; broken or loose wire; burnt-out solenoid coil; spool stuck because of dirt or varnish; no hydraulic pressure / pilot pressure (for pilot-operated valves); wrong voltage on the coil. (4)

5.4 Any FIVE ✓✓✓✓✓: dirty oil added when topping up; open filler cap or damaged breather; dirt entering through worn rod seals and wipers; metal particles from wear inside pumps and valves; dirt let in when components or hoses are replaced without plugging the ports; condensation (water) in the tank; blocked or damaged filters; new components that were not cleaned. (5)

[20]

TOTAL: 100

END OF MARKING GUIDELINE